

Proof Of Conjecture 2.3

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Notation and known identities

Let

$$\alpha = \theta_2 - \theta_4, \quad \beta = \theta_3 - \theta_4, \quad (1)$$

with $0 < \beta < \alpha < 2\pi$ (so $\theta_2 > \theta_3 > \theta_4$).

Define

$$p_{24} = \cos \alpha, \quad p_{34} = \cos \beta, \quad p_{23} = \cos(\alpha - \beta). \quad (2)$$

From Lemma 2.2 of the paper we have the trigonometric identities

$$1 - p_{24} - p_{34} + p_{23} = 4 \sin \frac{\alpha}{2} \sin \frac{\beta}{2} \cos \frac{\alpha - \beta}{2}, \quad (3)$$

$$1 - p_{24} = 2 \sin^2 \frac{\alpha}{2}, \quad (4)$$

$$1 - p_{34} = 2 \sin^2 \frac{\beta}{2}. \quad (5)$$

Hence the constant

$$c = \frac{1 - p_{24} - p_{34} + p_{23}}{2\sqrt{(1 - p_{24})(1 - p_{34})}} = \cos \frac{\alpha - \beta}{2} (> 0). \quad (6)$$

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Unsquared equation

The original geometric condition (equation (15) in the paper) is, with the sign chosen so that both sides are positive,

$$\frac{x^2 - x(p_{24} + p_{34}) + p_{23}}{\sqrt{(x^2 + 1 - 2xp_{24})(x^2 + 1 - 2xp_{34})}} = c. \quad (7)$$

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Squaring and clearing denominators

Squaring (7) and multiplying by the denominator gives

$$c^2 (x^2 + 1 - 2xp_{24})(x^2 + 1 - 2xp_{34}) = (x^2 - x(p_{24} + p_{34}) + p_{23})^2. \quad (8)$$

Substitute $p_{24} = \cos \alpha$, $p_{34} = \cos \beta$, $p_{23} = \cos(\alpha - \beta)$ and define

$$P(x) = c^2 (x^2 + 1 - 2x \cos \alpha)(x^2 + 1 - 2x \cos \beta) - (x^2 - x(\cos \alpha + \cos \beta) + \cos(\alpha - \beta))^2. \quad (9)$$

Then equation (8) is exactly $P(x) = 0$.

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Verifying that $x = \pm 1$ are roots

For $x = 1$:

$$x^2 + 1 - 2x \cos \alpha = 2 - 2 \cos \alpha = 4 \sin^2 \frac{\alpha}{2}, \quad (10)$$

$$x^2 + 1 - 2x \cos \beta = 2 - 2 \cos \beta = 4 \sin^2 \frac{\beta}{2}. \quad (11)$$

The first term of $P(1)$ becomes

$$c^2 \cdot 4 \sin^2 \frac{\alpha}{2} \cdot 4 \sin^2 \frac{\beta}{2} = 16 c^2 \sin^2 \frac{\alpha}{2} \sin^2 \frac{\beta}{2}. \quad (12)$$

The second term uses

$$1 - (\cos \alpha + \cos \beta) + \cos(\alpha - \beta) = 1 - p_{24} - p_{34} + p_{23} = 4 \sin \frac{\alpha}{2} \sin \frac{\beta}{2} c, \quad (13)$$

by (3) and (6). Squaring gives

$$(4 \sin \frac{\alpha}{2} \sin \frac{\beta}{2} c)^2 = 16 c^2 \sin^2 \frac{\alpha}{2} \sin^2 \frac{\beta}{2}. \quad (14)$$

Thus $P(1) = 0$.

For $x = -1$:

$$x^2 + 1 - 2x \cos \alpha = 1 + 1 + 2 \cos \alpha = 2 + 2 \cos \alpha = 4 \cos^2 \frac{\alpha}{2}, \quad (15)$$

$$x^2 + 1 - 2x \cos \beta = 1 + 1 + 2 \cos \beta = 2 + 2 \cos \beta = 4 \cos^2 \frac{\beta}{2}. \quad (16)$$

The first term of $P(-1)$ becomes

$$c^2 \cdot 4 \cos^2 \frac{\alpha}{2} \cdot 4 \cos^2 \frac{\beta}{2} = 16 c^2 \cos^2 \frac{\alpha}{2} \cos^2 \frac{\beta}{2}. \quad (17)$$

The second term:

$$1 + (\cos \alpha + \cos \beta) + \cos(\alpha - \beta) = 1 + \cos \alpha + \cos \beta + \cos(\alpha - \beta). \quad (18)$$

Using $\cos \alpha = 2 \cos^2 \frac{\alpha}{2} - 1$ and $\cos \beta = 2 \cos^2 \frac{\beta}{2} - 1$, we obtain

$$1 + \cos \alpha + \cos \beta + \cos(\alpha - \beta) = 2 \cos^2 \frac{\alpha}{2} + 2 \cos^2 \frac{\beta}{2} + \cos(\alpha - \beta) - 1. \quad (19)$$

A more compact identity is

$$1 + \cos \alpha + \cos \beta + \cos(\alpha - \beta) = 4 \cos \frac{\alpha}{2} \cos \frac{\beta}{2} \cos \frac{\alpha - \beta}{2}, \quad (20)$$

which can be verified by standard product-to-sum formulas. Since $\cos \frac{\alpha - \beta}{2} = c$, (20) equals

$$4 \cos \frac{\alpha}{2} \cos \frac{\beta}{2} c. \quad (21)$$

Squaring gives

$$(4 \cos \frac{\alpha}{2} \cos \frac{\beta}{2} c)^2 = 16 c^2 \cos^2 \frac{\alpha}{2} \cos^2 \frac{\beta}{2}, \quad (22)$$

identical to (17). Hence $P(-1) = 0$.

Therefore $x = 1$ and $x = -1$ are roots of $P(x)$, so $(x^2 - 1)$ divides $P(x)$. Consequently

$$P(x) = (x^2 - 1) Q(x), \quad (23)$$

where $Q(x)$ is a quadratic polynomial.

Explicit form of $Q(x)$

By expanding $P(x)$ and factoring out $(x^2 - 1)$ (or by using the trigonometric identities from Lemma 2.3 of the paper), one obtains

$$Q(x) = 4 \sin^2 \frac{\alpha}{2} \sin^2 \frac{\beta}{2} (x - 1)^2 + 4 \left(\sin \frac{\alpha}{2} \cos \frac{\beta}{2} - \sin \frac{\beta}{2} \cos \frac{\alpha}{2} \right)^2 x. \quad (24)$$

Observe that

$$\sin \frac{\alpha}{2} \cos \frac{\beta}{2} - \sin \frac{\beta}{2} \cos \frac{\alpha}{2} = \sin \frac{\alpha - \beta}{2} \neq 0 \quad (\text{since } \alpha \neq \beta). \quad (25)$$

Thus the second term in (24) is

$$4 \sin^2 \frac{\alpha - \beta}{2} x = 4(1 - c^2) x, \quad (26)$$

which is strictly positive for any $x > 0$. The first term is non-negative (it is a square times a positive constant). Hence

$$Q(x) > 0 \quad \text{for all } x > 0. \quad (27)$$

Positive roots of $P(x) = 0$

From (23) and (27), for $x > 0$ we have

$$P(x) = 0 \iff (x^2 - 1)Q(x) = 0. \quad (28)$$

Since $Q(x) > 0$ for $x > 0$, the only factor that can vanish is $x^2 - 1$. The positive root of $x^2 - 1 = 0$ is $x = 1$. Therefore

$$\text{The only positive solution of } P(x) = 0 \text{ is } x = 1. \quad (29)$$

Returning to the unsquared equation

Any solution of the original unsquared equation (7) automatically satisfies (8) and hence $P(x) = 0$. Moreover, for $x = 1$ the left-hand side of (7) equals c (positive), so $x = 1$ is indeed a solution. If there were another positive number $x_0 \neq 1$ satisfying (7), it would also satisfy $P(x_0) = 0$, contradicting (29). Thus

$$x = 1 \text{ is the unique positive solution of equation (7).} \quad (30)$$

Conclusion

The algebraic factorization $P(x) = (x^2 - 1)Q(x)$ with $Q(x) > 0$ for $x > 0$ proves that the quartic $P(x) = 0$ has exactly one positive root, namely $x = 1$. This establishes Conjecture 2.3 and consequently Corollary 2.3: the only positive solution of the original geometric condition is $x = 1$.

The constant c appears implicitly in the expression of $Q(x)$ through the relation $c = \cos \frac{\alpha - \beta}{2}$ (e.g., the second term equals $4(1 - c^2)x$), but the crucial property $Q(x) > 0$ for $x > 0$ holds independently.